Supplementary Protocol: Temporal and Evidence-Source Provenance-Aware Multimodal Misinformation Detection

Sabbir Hossain
American International University-Bangladesh
26-94030-1@student.aiub.edu

1 Purpose and Evidence Status

This supplement records the operational protocol before real-data experiments. It distinguishes proposed analyses from measured findings. The included smoke-test outputs validate software flow only and must not be cited as scientific results.

2 Eligibility and Rating Mapping

Eligible records are English image–text claims concerning United States events from 2020–2026. A record requires an identifiable claim time, an archived or recoverable image, an explicit professional rating, and enough metadata for temporal filtering. Only unequivocal ratings—true or correct; false, fake, or miscaptioned; and unproven, unsubstantiated, or no evidence—are mapped. Mixed, partially true, satire, opinion, and ambiguous ratings are excluded.

3 Temporal and Group Splits

Training uses 2020–2023 records, validation uses 2024, and the final test uses 2025–2026. The same event, near-duplicate claim, and perceptually identical image must not cross partitions. All preprocessing and thresholds are fitted using training or validation data only.

4 Planned Configuration

Component	Planned setting
Evidence discovery	ClaimReview, GDELT, and Common Crawl metadata
Text retrieval	BM25 and BGE-M3
Visual retrieval	SigLIP2 and perceptual hashing
Fusion	Reciprocal-rank fusion followed by reranking
Candidate limits	Rerank 100; retain 20 time-valid items
Graph	Heterogeneous temporal-provenance graph
Decision	Sufficiency gate followed by binary veracity head
Calibration	Validation-fitted temperature scaling
Abstention	Validation threshold targeting 80% coverage
Repetitions	Five seeds after model selection
Uncertainty	10,000 paired event-level bootstrap replicates

5 Ablations and Stress Tests

The preregistered ablations remove temporal filtering, dependency clustering, temporal graph features, graph aggregation, textual evidence, visual evidence, the sufficiency gate, and evidence-dropout training. Stress tests inject syndicated copies, add post-cutoff verdict-bearing documents, remove the earliest source, introduce evidence conflicts, swap captions, and reuse historical images.

6 Dataset Release Policy

The intended release contains code, URL and hash metadata, derived graph annotations, retrieval logs, and split manifests. Copyrighted article text and images are not redistributed without explicit permission. Fact-check verdicts remain inherited labels, and automatically extracted evidence links are treated as silver evidence.

7 Reporting Checklist

The final submission must report the data snapshot, model and tokenizer revisions, prompts, hardware, precision, search budget, seeds, failures, selection rule, calibration data, decision threshold, elapsed time, and raw prediction locations. Every table must be reproducible from preserved result artifacts.